

How Regulation Sentiments Influence Cryptocurrency? Is Crypto a Wall Street Darling?

Yosef Bonaparte
University of Colorado Denver
1475 Lawrence St, Denver, CO 80202
yosef.bonaparte@ucdenver.edu

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Abstract

This paper analyses the prospect of cryptocurrency regulation, especially, on crypto prices, volatility and volume. We employ the Google Trend technique and introduce the Crypto Regulation Sentiment Index (CRSX) that reflects Crypto regulation fear at the US and Global levels. We utilize over 75% of the crypto financial market daily data, including Bitcoin and several Altcoins (Ethereum, Solana, etc.). Surprisingly we find that CRSX has small and not statistically significant long term impact on cryptocurrency prices, and thus we classify crypto as the “Wall street Darling,” as a reflection of the forgiveness that investors exhibit to crypto bad news. Yet, CRSX exhibit a large impact on cryptocurrency volatility and trading volume, especially in recent years, perhaps due to the fact that crypto becomes more as a main stream. Moreover; the CRSX impact on crypto varies by coin’s key blockchain characteristics, especially, the type of blockchain consensus mechanism (POF vs POS). Collectively, we demonstrate that the regulating sentiment impacts crypto volatility and volume and varies by the type of the employed consensus mechanism.

Keywords: Consensus Mechanism, blockchain finance, Regulation sentiment, Proof of Work (POF), Proof-of-Stake (POS), Ethereum and Bitcoin.

JEL Classification: G10, G12

1. Introduction

The Cryptocurrency market exceeds the \$2 trillion market cap mark as of 4/2022, where Bitcoin stands at 878 billion and the Altcoin over \$1.2 trillion; and Ethereum around \$390 Billion. In fact, the crypto asset class is fast growing asset class; a report by the JPMorgan suggests that Bitcoin price could rise over \$146K in the long term.¹ While most work focus on the key aggregate determinants of cryptocurrency prices, e.g. Biais et. al. (2018), or focusing one short term (few days post event) single regulation event, Auer, R. and S. Claessens (2018); this paper zooms on the regulation sentiment and analyses how regulations may impact cryptocurrency overtime with long-term prospective and analyzing various aspects, including prices, volatility and trading volume.

Indeed, our goal is to examine the economic significant of regulation on key crypto finance performances or outcomes. Such an analyses faces deep econometrics challenges: while we have time series data about crypto performance (prices, volatility and volume), we don't have a time series data about regulations overtime. Of course, we can analyze the impact of new regulation for a few days, yet our goals are beyond short term analyses and focus more on long term analyses.

To address this critical econometrics problem, we set the stage by first composing two regulation sentiment indexes, one for the US and the other one is globally. We use the Google Trends search engine that analyses subject search volume intensity, similar to the approach performed by Dimpfl and Jank (2016), Vlastakis and Markellos (2012), Vozlyublennaiia (2014) and Da et al., (2011). Specifically, the Google Trends track attention to several terms related to cryptocurrency, banning, regulation, mining and illegal. Several studies document that Google's search engine is the leading search tool, used for 77.8% of searches among the different search engines on the web.²

¹ Article URL: <https://www.coindesk.com/jpmorgan-predicts-bitcoin-price-could-rise-over-146000-in-long-term>

² For instance, see URL: <https://www.netmarketshare.com/search-engine-market-share.aspx?qprid=4&qpcustomid=0>

After we compose the US and Global CRSX, then we conduct time series analyses and find that there is no large differences between the two, and thus we only use the Global CRSX for the rest of the analyses. Furthermore, we find that CRSX is serially correlated, which means this month regulation sentiments influence the following month sentiment, as such, we consider spillover in our analyses (forward 1 and 2 months). The first step of our analyses is collect daily data about cryptocurrencies, Bitcoin and altcoins (Ethereum, Doge, BNB, Cardano, etc.). This data sets information generate monthly data about crypto prices and returns as well as on crypto volatility and volume. For each coin, we also consider control variables that used in these types of regressions, e.g., Bonaparte (2022), and collect coin's key data features about: coin's market cap, age (trading period) and the type of consensus mechanism (one if proof-of-work, 0 otherwise).

After collecting the CRSX and data about cryptocurrencies, we then turn and examine the impact of CRSX on crypto for a host of financial outcomes: prices, volatility and volume. In our econometrics strategy, if there is no linear relationship between CRSX and a crypto financial outcome, we then try non-linear specification as well as examining concurrent, one month and two months forward (possible spillover). Furthermore, since some coins are older than other coins, in terms of trading history and Bitcoin is the oldest, we employ coin cluster regressions as well as employing years and months fixed effects.

Surprisingly, our results demonstrate that CRSX has little impact on crypto prices at the concurrent, and slightly but not statistically significant on the one and two months forward. While the short term analyses demonstrate a drop in the price when regulations appear, e.g., Auer et. al. (2018), perhaps such a regulation news grab attention of investors and buy more, e.g., Barber and Odean (2008); so it offset the negative news. We name this phenomena as Wall Street Darling that

reflects when an asset has bad news, still investors continue to hold it, being forgiving and support its price.³ We observe similar behavior with Elon Musk of Tesla and at some point with Meta (Facebook).

On the other hand, CRSX has positive and statistically significant influence on crypto volatility and log volume. Furthermore, we employ an interaction term between CRSX and each one of the coin's control variables (age, market cap and consensus mechanism) and find that CRSX impact on crypto financial outcomes varies by coin's these key characteristics. Besides, we employ subsamples analyses before and after January of 2020; this is the time that nicely separates our sample into two fairly equal observations, return volatility and volume, and also reflect the period before and after Covid. We find that CRSX has greater impact post 1/2020 and perhaps this because cryptocurrency becomes more and more as mainstream.⁴

This paper is important for at least four main contributions. First, to the best of our knowledge, this is the first paper that generates a time series regulation sentiment index that aids us, researchers and policy makers as well as the business community, to measure the overall sensitivity of crypto to regulations. Second, the findings of our research are dramatic for which we demonstrate that regulation has no big impact on prices, it is more of volatility and volume, which possibly suggest that regulations already is priced. Third, we show that the more cryptocurrency becomes a main stream the greater the impact of regulations on volatility and trading volume. Fourth, we present a formal definition with econometrics presentation as well as an application to a behavioral finance concept, Wall Street Darling to the literature. We believe these additions enhance our understanding when we express investors being forgiving for an asset that its price does not drop despite bad news.

³ The term Wall Street Darling used widely, here's an example in Investopedia article <https://www.investopedia.com/updates/enron-scandal-summary/>

⁴ See ABC TV <https://abcnews.go.com/US/wireStory/bitcoin-mainstream-wall-street-cash-80663635#:~:text=NEW%20YORK%20%2D%2D%20Love%20cryptocurrencies,more%20than%20200%20million%20users> and Yahoo Finance <https://finance.yahoo.com/news/crypto-will-become-mainstream-says-most-professional-investors-study-190006088.html>

Collectively, we believe this paper demonstrates that regulations has impact on key crypto financial indicators, including crypto volatility and volume; and the impact varies by key coin's features and greater post Covid.

1.2 Regulation: threat versus opportunity

This subsection reviews the key regulatory stop in the crypto ecosystem and what are the opportunities and the threat in this domain. First, it is important to identify the key players who may be involved in crypto regulation:

1. Governments
2. Federal Reserves or Central Banks
3. Security and Exchange Commission

In every country, the government and its parliament pass regulations in many directions. Such as banning or allowing mining, impose taxations on crypto capital gain, allowing transaction with cryptocurrency, etc. It is important to understand why governments needs regulations? Some to protect against possible frauds, also to defend local currency from being second used currency, and to gain some revenue from this business.

Now we turn to federal reserves or central banks; they also can play a role in the ecosystem of crypto regulation via regulating what banks can use for reserve or prevent banks from using cryptocurrency for banking activities. Finally, the Security Exchange Commissions (SEC) can regulate the exchanges and what that will allow/prevent investors, especially institutional investors, from trading cryptocurrency in their businesses.

Next, we turn to analyze the opportunities and threat from regulating cryptocurrency. We first map the opportunities, and we list at least three main advantages:

1. Protection against fraud and deceptions

2. Bring stability and lower the cryptocurrency volatility
3. Add more confidence to investors and better ecosystem environment
4. More investments from institutional investors.

On the other hand, the disadvantage from regulating cryptocurrency are:

1. Obstruct innovation and technological progress
2. In contradiction of the decentralization spirit of cryptocurrency as its core emphasis
3. Trading opportunities: price stability cause lower volatility reduce the trading volume

What is the next step in crypto regulation?

Recently (early 2022), President Biden signed off on new crypto tax legislation along with the \$1.2 trillion bipartisan infrastructure bill, that passed in 2021, while the Federal Reserve is considering the idea of issuing a U.S. digital currency. In fact, the Chairman of the Federal Reserve Jerome Powell talked on July 2021 to congress about the Federal Reserve's interest in regulating stable-coins, and the potential for a **Central Bank Digital Currency (CBDC)**; A CBDC would make it easier to make transactions digitally. Now, all eyes are on a coming report from the Federal Reserve, which Powell expects to publish around early September; where the Federal Reserve is going to address digital payments broadly.

Along these new polices, the SEC outlines a new set of cryptocurrency accounting standards in order to protect crypto assets held by companies, especially, for users against hacking losses. This development move originates because of more trading platforms permits users to do business with crypto and the hacks continue to occur. To sum up, the advantage of regulating cryptocurrency is mainly present a safer crypto ecosystem while the disadvantage is obstruction to innovation and contradict the decentralize finance spirit. We believe there is a long way to go till we reach that point, at least in the United States, where we can claim that the cryptocurrency ecosystem is regulated.

The rest of the paper is organized as follows. Section 2 reports the data sources and reviews the related literature. Section 3 reports the econometrics method to compose our CRSX index and then presents the econometrics models that we employ in our analyses. Section 4 reports the results and analyses inference; and Section 5 summarizes our findings and draws the conclusion that CRSX impact key crypto financial indicators.

2. Data Sources and Literature Review

This section first reports the literature review for studies related to our work, so to enhance the contribution and the added value of our paper to the broad range of the literature. We then report the data set that we utilize including sources; and then define key variables and present summary statistics of the variables as well as key coin's characteristics.

2.1 Literature Review

This subsection reviews the literature that is related to regulations and cryptocurrency and then reports the studies that support our methodology to compose the crypto regulation index. Indeed, the cryptocurrency literature is extensive and spread, ranging from theoretical to empirical analyses; from physical crypto mining to blockchain technology and computer power. Our goal is to summarize it, so to infer our added value of our study to the overall literature.

We begin the literature review with theoretical side, namely, studies by Biais, Bisiere, Bouvard, Casamatta, and Menkveld (2018) and Pagnotta and Buraschi (2018) suggest that blockchain actions may capture the dependability and transaction benefits of a blockchain and that indicate the cryptocurrency values is determined by computing power and network size of blockchains. Additionally, the work by Saleh (2018) suggests that the relationship between computing power and electricity consumption is non-negative and even positive; in this line of work a work by Sockin and

Xiong (2018) demonstrate how “trustless” aspect of decentralized networks is a significant part for innovation of blockchain technology, which depicts some inference of the value of the blockchain.

A study by Makarov and Schoar (2019) demonstrate that the price-deviation events may affect cryptocurrency returns. There is a line of work that links between cryptocurrency prices and computing power, e.g., Pagnotta and Buraschi (2018). On the other hand, several studies, e.g., Abadi and Brunnermeier (2018), suggest that decentralized ledgers may cause some instability due to new ledgers that can divided off an existing blockchain.

Furthermore, a work by Catalini and Gans (2016) analyzes the advance of crypto throughout the year 2017 and propose that disruptive abilities of the blockchain technology may be one of the key determinants. Additionally, some studies focus on the mining blocks that later on form the blockchain; e.g., Nakamoto (2008); Narayanan, Bonneau, Felten, Miller, and Goldfeder (2016). Along this line of research a work by Cong, Li, and Wang (2018) examines the economics of non-mineable currencies, which they mention to as “tokens.”

Now we turn and focus on Bitcoin, as one of the largest market cap among cryptocurrency, and show that there is a great deal of discussion in the literature on if Bitcoin is a proper currency. We start with the work by Yermack (2015), which suggest that Bitcoin is a speculative asset since of relatively high price volatility as well as not being “yet” a proper currency. Along this line of research, Cheah and Fry (2015) contend that Bitcoin has no fundamental value, thus the price is mainly driven by investor sentiment; their research analyzes the period between 7/2010 and 7/2014 and displays that Bitcoin faced a number of “bubble-like” periods. A work by Auer and Claessens (2018) suggest a host of financial friction, such as market segmentation and regulatory news in a jurisdiction influence the prices of cryptocurrencies traded in that jurisdiction. Furthermore, a work by Foley, Karlsen, and Putnins (2019) gauge the period between 2009 and 2017 and show that approximately 46% of Bitcoin’s trading volume was related to illegal activities. A work by Biais et al. (2018) suggests an overlapping

generation's model with cryptocurrency, where the model includes monetary purpose instead of portfolio choice model.

We continue on Bitcoin and review the counter side in the literature. Namely, a work by Pagano and Sedunov (2019) propose that Bitcoin is a true form of money and medium of exchange especially, in recent years, where more vendors agree to use it. Besides, Selgin (2015) and Baur et al. (2018) contend that Bitcoin syndicates characteristics of commodity money and fiat currencies. Additionally, Makarov and Schoar (2019) advocate that the cryptocurrency market is subject to arbitrage profit because prices vary through exchanges for weeks; Bitcoin, Ethereum, and Ripple were included in the analyses. Some work show how crypto may facilitate some economic cumbersome; e.g., Cong and He (2019) demonstrate that blockchain technology able to assist executing contracts automatically. Yermack (2017) advocates that blockchain practice may advance corporate governance. Chiu and Koepl (2019) suggest blockchain technology may advance the settlement of securities.

Moreover, a work by Baur et al. (2018) demonstrate that indeed Bitcoin is uncorrelated to stock equity indexes, bonds, and commodities; a study by Liu and Tsyvinski (2018 and 2019) echo these findings and show that Bitcoin and some altcoins like Ripple and Ethereum also are unrelated to equity, bonds, currencies, and even precious metals markets. Some studies zoomed on microstructure issues; such as, a study by Easley, O'Hara, and Basu (2019) display that the transaction fees paid to miners for the development of the Bitcoin market are commanding to maintain the blockchain feasible, especially as more and more blocks are being mined. Along this line of work, Bonaparte (2021 A and B) demonstrate how time horizon influence the decision to own crypto asset class, and introduces a newly crypto VIX named as CVIX the reflects the 30 days forward of crypto volatility.

Finally, several studies, e.g. Cong et al. (2018), Houy (2014), Malinova and Park (2017), Cong and He (2019) and Iyidogan (2019), Tinn (2017), Huberman, Leshno, and Moallemi (2017), Li, Ma,

and Chang (2018), Prat and Walter (2018), Chiu and Koepl (2019), examine the significance of a host of key factors that links between functioning of blockchain and computer power, adoption levels, and transaction fees.

Next, we turn to review studies that use Google Trends data in their work, in order to show the validity of Google Trends as a tool for investor news priorities. We use the Google Trends tool to support our hypotheses and inference about the impact of crypto regulation.

Studies by Vlastakis and Markellos (2012), Vozlyublennai (2014) and Dimpfl and Jank (2016) suggest that Google searches function as a good proxy for retail investor news attention. Other papers demonstrate that search intensity varies over time; for example, Dzielinski (2012) shows that investors intensify their search in response to greater uncertainty; and Vozlyublennai (2014) shows that more stock related Google searches are conducted when stock prices decline. The search intensity is also studied in economic psychology literature; Lemieux and Peterson (2011), and Abbas et al. (2013), provide empirical evidence that individuals intensify their search activity in response to greater price uncertainty.

A related line of research demonstrates that Google Search Volume Index (SVI) data provides information that forecasts near-term economic activities, such as automotive sales, tourism, and home sales, Choi and Varian (2012). SVI data can also be used to predict diffusion phenomena such as flu outbreaks, Ginsberg et al. (2009). The study that is most related to our work is Da et al. (2011), who show that search volume traffic in Google search engines can be a direct measure of investor attention.

Following these insights, we use Google SVI to track internet news search volume over time, using this as a proxy for investor news priorities. Google is the leading search engine, used for 77.8% of searches among all search engines, so we use it to test our hypotheses because it covers the majority of individuals.

2.2 Data sources and variable definitions

In this subsection we report the data sets we utilize, including the definition of key variables. We collect three data sets and from different resources. The first set of data is about composing the CRSX from the Google trend. Our method to compose the sentiment index related to crypto regulation is by using monthly data from Google Trends to obtain search volume for keyword phrases commonly associated with crypto and regulation. We first conduct research to identify the most commonly used keyword phrases, we then use Google SVI as our proxies of crypto regulation sentiment CRSX. To capture CRSX, we find that, among the terms extensively used by browsers to follow crypto regulation activity, the keyword phrases: “bitcoin regulation,” “bitcoin regulation”, “cryptocurrency regulation”, “cryptocurrency ban”, “bitcoin ban”, “bitcoin mining illegal”, “crypto mining illegal” are the most suitable ones. We find that “bitcoin mining illegal” is the most used term followed by “cryptocurrency regulation” and then “bitcoin ban.”

Second, the data for each coin daily prices and volume is from Yahoo Finance, which gather directly from the exchange open source. We collect data from different coins, including: Bitcoin and Altcoin (Ethereum, BNB, Dogecoin, Solano and Cardano); the market cap of these coins nearing \$1.4 trillion that is the wide majority of the market cap, which nears \$2 trillion. For each coin, we use the closing daily prices to calculate the daily return. Since our analyses is monthly, we use the daily return to calculate the monthly return, and then use the standard deviation of the daily return to measure the monthly volatility. In addition, we sum the daily trading volume and create the monthly sum volume. We conduct the same approach to each coin, as such, we now have data about the crypto financial indicators, the right hand side of our regression model.

The third set of data is about each coin’s key characteristics. We collect three main features by each coin as follows. First, the age (trading history) and this is also can be inferred using Yahoo Finance data. Second, market cap for each coin and we use the data from <https://coinmarketcap.com/> that

reports the current market cap for each coin; as of the date of this study (3/2022). Using the monthly return, we can track back the market cap for each coin by month and year. Third, the consensus mechanism for each coin. Here we create a binary variable that takes a value of 1 if the consensus mechanism is a proof-of-work (POF) and 0 otherwise. In our sample, most coins employ POF consensus mechanism, especially, Bitcoin, Ethereum (will be change later on 2022) and Dogecoin, while the other coins in the sample have a different consensus mechanism, proof-of-stake. Bonaparte (2022) demonstrate that these three characteristics are the main ones when we analyze crypto financial indicators.

Table 1 reports the key feature of each coin that we analyze in this study, namely, the age market cap and consensus mechanism, with Bitcoin and Ethereum are the largest coin and using POW and with market cap of \$810 and \$390 billion, respectively. Table 2 reports the summary statistics of the key variables, and it shows that on average US CRSX is 90 with a standard deviation of 93; while the average of global CRSX (divided by 100 for normalization) is 1.126 and with standard deviation of 1.067. The average return and volatility of crypto is 3.13% and 0.05%, respectively, while the log volume stands at 23.8 with a standard deviation of 2.55. The average age of coins is 37 months and average market cap is 207 billion and 63% of the observation with a proof-of-work consensus mechanism. Finally, in Appendix A we report the definitions of key variables that we utilize in this study.

3. Composing CRSX and Econometrics Models

This section presents our econometrics methodology, specifically we displays two main subsections; one about our method to construct our cryptocurrency regulation sentiment index, CRSX, and then we present our econometrics models that we employ to estimate our models to gather inference.

3.1 Cryptocurrency Regulation Sentiment Index - CRSX

This subsection key time series analyses on the global CRSX as well as on the US CRSX. We conduct the Google SVI on the US and on the entire world to capture the global. We plot the Google search volume intensity for these terms for the US and the global in Figure 1, which depicts the CRSX between 1/2004 and 3/2022. The blue line measures the Global CRSX while the black curve reports the US CRSX. The figure demonstrates some seasonality or peaking around the 2017 and 2018 as well as on the period of 2020. Between these two peaks, we also noticing a permanent rise of CRSX than before 2017. Moreover, we observe that there is not a big large difference between the US CRSX and the Global CRSX, as both demonstrate similar patterns.

We further plot the global CRSX over the price of Bitcoin in figure 2, which depicts the CRSX the normalized price of Bitcoin between 1/2014 and 3/2022. The dark blue line measures the price of Bitcoin whereas the light blue depicts the Global CRSX. The figure demonstrate some sensitivity between CRSX and Bitcoin price around the year 2017 but less in the second peak of CRSX post 2020. Altogether, our analyses demonstrate that the US and the global CRSX exhibit similar pattern and not large differences; and that CRSX is seasonal and the price of bitcoin is not very sensitive in the second peak.

Next, we conduct a time series analyses of the CRSX, both the US and the global and employ the following model:

$$(1) \quad CRSX_t = c + \rho * CRSX_{t-1} + \varepsilon_t$$

Here $CRSX_t$ is the crypto regulation sentiment index at time (year and month) t , c is a constant and ρ is the serial correlation, which reflects the comovement between the current and the future; ε_t is the residual terms and reflects the unobservable.

We report the results of the model in table 3, for which the dependent variable in regressions (1) and (2) is global CRSX and the key independent variable is the US CRSX. The dependent variable in regression (3) is global CRSX and the independent variable is lag CRSX. In regressions (1) and (3) we omitted the constant. The dependent variable in regression (4) is the US CRSX and the key independent variable is lag US CRSX. The number in parentheses measures the t -statistics and we employ OLS regression with robust standard error. Definition of key variables are in Appendix A.

The results demonstrate two main inferences. First, from the results of regressions (1) and (2) show that the US and the Global CRSX are highly comove and correlated, with an R-square of 0.95 near perfect. Second, the results of regressions (3) and (4) demonstrate that the serial correlation for both the US and the global CRSX is near $\rho = 0.971 > 0$, which means both the US and the global CRSX are positively autocorrelated; that reflects today CRSX sentiment positively influences future CRSX sentiment. As such, we consider spillover analyses when we examine the impact of CRSX on crypto.

To sum up, the time series analyses of CRSX demonstrate that the US and the global are highly commove with each other and there positive serial correlation, that reflects the momentum in the crypto regulation sentiment.

3.2 Econometrics model

This section analyses the impact of regulations on key crypto financial indicators, namely, return, volatility and volume. The econometrics model that we employ the following model:

$$(2) \quad \text{Crypto Financial Outcomes}_t = c + \beta * X_t + \delta * \text{CRSX}_t + \varepsilon_t$$

Here the *Crypto Financial Outcomes_t* is the crypto financial outcomes at time t , and includes three indicators: the crypto return denoted as *Crypto Return_t*, the crypto volatility denoted as *Crypto Volatility_t* and the crypto volume denoted as *Crypto Volume_t*. The coefficient c is a constant and β is a set of coefficients for the set of coin's control variables X_t , which includes coin's age, market cap and the consensus mechanism. The key quantity is our regulation index denoted as *CRSX_t* and the sign of the coefficient δ is our key estimate as it reflects the overall impact of regulation sentiments on key financial indicators. Specifically, a positive $\delta > 0$ means CRSX positively impact crypto prices or volatility or volume; whereas negative $\delta < 0$ means the opposite. If $\delta = 0$ and statistically significant, it suggests that CRSX has no impact on key crypto financial indicators.

It is important to mention that we also consider to employ a non-linear specification in case and we find that the key coefficient $\delta = 0$ or near zero and not statistically significant, hence,

$$(3) \quad \text{Crypto Financial Indicator}_t = c + \beta * X_t + \delta * \text{CRSX}_t + \theta * \text{CRSX}_t^2 + \varepsilon_t$$

Here the coefficient measures if there is a non-linear relationship on the impact of CRSX on key crypto financial performance. A positive $\theta > 0$ reflects a convex relationship, whereas $\theta < 0$ a concave relationship, and $\theta = 0$ means no quadratic relationship.

Moreover, we consider spillover on the impact of CRSX, for which the crypto regulation sentiment not only may have impact on the concurrent, but also on future crypto financial performances, as such, we employ the following model:

$$(4) \quad \text{Crypto Financial Indicator}_{t+j} = c + \beta * X_t + \delta * \text{CRSX}_t + \varepsilon_t$$

The key difference between equation (3) and (2) is the sub time indicator j where when $j = 0$ then equation (3) is identical to equation (2), while when $j > 0$ it means we examine the spillover of the impact of CRSX on key crypto financial performances. In our empirical analyses we consider only two months spillovers, hence, $j = 1$ and $j = 2$.

We believe considering the nonlinear relationship with the quantity $\theta * CRSX_t^2$ as well as considering analyzing spillover with $j = 1, 2$ add some flexibility to the analyses and enrich our understanding on the overall impact of CRSX on key crypto financial outcomes. We also conduct interaction terms between $CRSX_t$ and X_t to gather some inference on if the impact of $CRSX_t$ varies by some coin key characteristics, such as, market cap and age; which will aid us to understand the influence of $CRSX_t$ on coin's return, volatility and volume.

In all of our empirical analyses we control for years and months fixed effects as well as clustering for coin because we don't want to have one coin to have greater weight than other just because it has greater history and more observations, e.g., Bitcoin and Dogecoin. We believe this treatment will ease some econometrics challenging or questioning.

4. Crypto Financial Outcomes and CRSX

This section reports the estimation results for which the key dependent variable is crypto currency financial outcomes, including, return, volatility and volume. We aim to analyze the impact of crypto regulation, proxy by CRSX, on crypto financial outcomes.

4.1 The Impact of Regulations on Crypto Return and Volatility

This subsection reports the baseline results, for which in all models that we estimate we consider non-linear relationship as well as the possible spillovers and interaction terms between CRSX and key coin's properties. The first model we employ is:

$$(5) \quad \text{Crypto Return}_{t+j} = c + \beta * X_t + \delta * \text{CRSX}_t + \theta * \text{CRSX}_t^2 + \varepsilon_t$$

Here the quantity $\text{Crypto Return}_{t+j}$ is the return of crypto at time $t + j$. We report the results in table 4 for which the dependent variable in regressions (1), (2) and (5) is crypto return at time t (concurrent) and in regressions (4) and (5) crypto return with one month forward and two months forward, respectively. The key independent variable is global CRSX and the market cap, trading age of the crypto as well as the type of the consensus mechanism (1 if POF and 0 otherwise). We employed three interaction terms with CRSX: with market age (CRSX*Age); market cap (CRSX*MC) and consensus mechanism (CRSX*CM). In all regressions, we control for year and month fixed effects, weighted average by coin cap and cluster by coin. The number in parentheses measures the t-statistics and we employ OLS regression with robust standard error. Definition of key variables are in Appendix A.

There are three sets of inference we may gather from estimating this model. First, the impact of CRSX on Crypto prices is not statistically significant; and there is no nonlinear relationship. Second, there is a negative and then positive spillover on the one month and two months forward, which means, when regulations risk arises, cause prices to go down then up as it grabs more attention and perhaps investors buy more of crypto, which is in line with Barber and Odean (2008), where they suggest that stocks in the news will have a net buys. Third, the interactions terms indicate that the impact of CRSX on crypto return varies by the key characteristics of the coin, age (less) and market

cap (more). As noted earlier, we name this surprising finding as a Wall Street Darling, in which there is a support for asset price despite the bad news; investors provide forgiveness or labeled as forgiving investors.

Next, we examine the impact of CRSX on Crypto volatility and employing the following model:

$$(6) \quad \text{Crypto Volatility}_{t+j} = c + \beta * X_t + \delta * \text{CRSX}_t + \theta * \text{CRSX}_t^2 + \varepsilon_t$$

Here the quantity *Crypto Volatility*_{*t+j*} is the crypto volatility at time *t + j*. We report the results in table 5 for which the dependent variable in regressions (1), (2) and (5) is crypto volatility at time *t* (concurrent) and in regressions (4) and (5) crypto volatility with one month forward and two months forward, respectively. The key independent variable is global CRSX and the market cap, trading age of the crypto as well as the type of the consensus mechanism (1 if POF and 0 otherwise). We employed three interaction terms with CRSX: with market age (CRSX*Age); market cap (CRSX*MC) and consensus mechanism (CRSX*CM). In all regressions, we control for year and month fixed effects, weighted average by coin cap and cluster by coin. The number in parentheses measures the *t*-statistics and we employ OLS regression with robust standard error. Definition of key variables are in Appendix A.

There are three sets of inference that we collect from estimating this model. First, the impact of CRSX on Crypto volatility is linear positive and statistically significant; yet there is no nonlinear relationship. Second, the spillover is weak and declining on the following months and becomes less statistically significant. Third, the interactions terms indicate that the impact of CRSX on crypto volatility is fewer on old companies and greater on the proof-of-work consensus mechanism.

4.2 The Impact of Regulations on Crypto volume and Subsample Analyses

This subsection analyses the impact of CRSX on another crypto outcome, which is crypto trading volume, and then turn to conduct some subsample analyses, so to examine if CRSX impact on crypto financial outcomes changes over time, by considering two sample before and after January 2020.

We start with the impact of CRSX on crypto trading volume; as such, we employ the following model:

$$(7) \quad \text{Crypto Log Volume}_{t+j} = c + \beta * X_t + \delta * \text{CRSX}_t + \theta * \text{CRSX}_t^2 + \varepsilon_t$$

Here the quantity *Crypto Log Volume*_{*t+j*} is the log volume of crypto at time *t + j*. We report the results in table 6 for which the dependent variable in regressions (1), (2) and (5) is crypto volatility at time *t* (concurrent) and in regressions (4) and (5) crypto volatility with one month forward and two months forward, respectively. The key independent variable is global CRSX and the market cap, trading age of the crypto as well as the type of the consensus mechanism (1 if POF and 0 otherwise). We employed three interaction terms with CRSX: with market age (CRSX*Age); market cap (CRSX*MC) and consensus mechanism (CRSX*CM). In all regressions, we control for year and month fixed effects, weighted average by coin cap and cluster by coin. The number in parentheses measures the *t*-statistics and we employ OLS regression with robust standard error. Definition of key variables are in Appendix A.

There are three sets of inference that we collect from estimating this model. First, there is a positive and linear impact of CRSX on Crypto volume and it is statistically significant. Moreover, there is a nonlinear concave relationship, which means the greater the sentiment the less the positive impact of CRSX on Crypto volume. Second, the spillover impact is declining on the following months and turns less statistically significant. Third, the interactions terms indicate that the impact of CRSX on

crypto volume is varies by key coin's feature, with fewer impact on: old coins, POW consensus mechanism and large cap.

Now we turn to the subsample analyses, which is the final set of analyses. The goal is to examine whether regulation sentiment exhibit greater or less impact in the recent years relative to early years. We select the cutoff January 2020 to split the sample is it equally split the sample across all the crypto financial indicators and it is approximately before and after Covid erupts (on December 2019 Covid hit China). Therefore, we employ the following two models:

$$(8) \quad \{Crypto\ Financial\ Indicator_t = c + \beta * X_t + \delta * CRSX_t + \varepsilon_t | t < 1/2020\}$$

and

$$(9) \quad \{Crypto\ Financial\ Indicator_t = c + \beta * X_t + \delta * CRSX_t + \varepsilon_t | t \geq 1/2020\}$$

Here the notations “ $| t < 1/2020$ ” and “ $| t \geq 1/2020$ ” indicate estimating the model for only the subsample before January 2020 (equation 8) and after and includes January 2020 (equation 9).

We report the results in table 7, where the dependent variable in regressions (1) and (2) is crypto return, while in regressions (3) and (4) is crypto volatility and in regressions (5) and (6) is crypto trading volume. The key independent variable is global CRSX, market cap, age and consensus mechanism of the crypto. In all regressions, we control for year and month fixed effects, weighted average by coin cap and cluster by coin. The number in parentheses measures the t -statistics and we employ OLS regression with robust standard error. Definition of key variables are in Appendix A.

There are two main sets of results to infer from the subsample analyses. First, the impact on CRSX on crypto prices or return is not statistically significant before and after January 2020. Second, the impact of CRSX on crypto volatility and volume is greater recently, hence, it is increasing as more

and more crypto becomes a main stream. Therefore, we infer that the impact of crypto regulation sentiment or CRSX is increasing on volatility and volume overtime.

4.3 Crypto the Wall Street Darling: Final Thought

In this subsection we focus on the “Wall Street Darling” concept that we introduced and develop. For individual stocks, there is the well-known phenomenon of “Wall Street darlings”, where investors exhibit herding behavior, following headlines and flocking to invest in so-called “Wall Street darlings.” Wall Street darlings are stocks that are reaching new highs, mirrored by enthusiastic and positive coverage by the analysts and the financial press of Wall Street. Investors buy darling stocks when the price is high, drawn in as stock price reaches dramatic highs and there is great exuberance among analysts and the popular press. The same concept of “Wall Street darlings” can be extended to asset classes – that is, investors are drawn to asset categories that are in the headlines as they reach new highs.

For example, there is the well-known phenomenon of sector rotation between growth stocks and value stocks- money moves from one asset class to the other as one asset class or the other (value or growth, for example) becomes the new Wall Street darling – both individual investors and institutional investor exhibit this behavior – this movement of investment money mirrors the headlines in the financial and popular press, as a particular asset class gains favor on Wall Street, reflecting rising prices/returns of that asset class. Thus, the phenomenon of Wall Street darlings is seen for both individual stocks, and entire asset classes, and now we see it in cryptocurrency asset class.

In the behavioral finance literature there are great and rich concepts that depicts response to an event, such as, overreaction or underreaction, corrections, etc. We add a new concept, the Wall Street

Darling, to shed a light on a phenomena, where investors do present forgiveness to bad news for multiple times; which one can also view it as a constant underreaction.

To enhance our understanding, we present the following econometrics model:

$$Price_{t+i} = c + \alpha * X_t + \rho_i * Event_t + \varepsilon_t$$

Here the quantity $Price_{t+j}$ reflect the darling asset price at time t ; and X_t is a set of control variables, where α is the set of coefficients for these controls; and ε_t is a residual term reflect the unobservable. The key variable is on the quantity $Event_t$ that reflect the even bad or good news; it can be binary (takes a value of 1 after the news break and 0 otherwise) or it can be continuous, like our CRSX in this paper. The main emphasis, yet, is on the coefficient ρ_i value for different sub-index values of i . In particular, the relationship between the forward price $Price_{t+i}$ and the $Event_t$ determines the darling relationship.

If during the bad news, the value of $\rho_i \geq 0$ at $i = 0$ or for small realization of i , then we call this as a Wall Street Darling, because the bad news evaporated quickly; recall that if the event is a bad news, then we expect that $\rho_i < 0$, yet if we observe that it is zero or positive, especially after a short term, then we turn on the wall street darling light; because we expect that it supposed to be negative. Therefore, the sub-index i is critical for understanding the darling concept; for which the shorter the sub-index i that makes coefficient $\rho_i = 0$ the more darling is the asset class or the individual stock.

5. Summary and Conclusion

The goal of this paper is to investigate how regulations may impact key crypto financial indicators or performance. To address this research question, we ought to gather time series on regulations and crypto financial indicators. While the crypto outcomes are widely available, the regulation data is not organized in a manner to conduct time series analyses. As such, the first step of this study is to create a crypto regulation sentiment index that reflects the overall sentiment in the global and the US market. We follow previous studies and use the Google search trend and compose an index that measures crypto regulation sentiment; and name them as US CRSX and Global CRSX, respectively.

After composing the CRSX, we compose key time series inference and find that the US and the Global CRSX are highly correlated and each positively serially correlated, which means the value of today CRSX impact the future realization of CRSX. These findings has two implications, one is we use the global CRSX for the entire analyses, and second we examine possible spillover.

We then analyses the impact of CRSX on crypto return, volatility and volume on the concurrent and on the future as well as if there is a linear or quadratics relationship between CRSX and crypto performance. We find that CRSX has a small impact on crypto prices and statistically not significant. We suggest the term of Wall Street Darling to Crypto since investors are forgiving bad and news and continue to support the price. Furthermore, CRSX positively impact crypto volatility and volume and this impacts varies by the coin key features, such as, age (trading history), market cap and the type of the consensus mechanism.

In addition, we conduct time subsample analyses and find that the influence of CRSX on crypto financial indicators increasing over time, especially post January of 2020. Collectively, this study exhibit new findings related to crypto performance and regulation, especially, regulation has little impact on prices, and present a newly index that capture the crypto regulation sentiment that can be used for future research and examination.

References

- Abadi, J. and M. Brunnermeier (2018). Blockchain economics. *NBER Working Paper No. 25407*
- Abbas, A. E., Bakir, N. O., Klutke, G. A., & Sun, Z. (2013). Effects of risk aversion on the value of information in two-action decision problems. *Decision Analysis*, 10(3), 257-275.
- Auer, R. and S. Claessens (2018). Regulating cryptocurrencies: Assessing market reactions. *BIS Quarterly Review September*.
- Barber, B.M. and Odean, T. (2008) All that Glitters: The Effect of Attention and News on the Buying Behavior of Individual and Institutional Investors. *The Review of Financial Studies*, 21, 785-818.
- Baur, D. G., K. Hong, and A. D. Lee (2018). Bitcoin: Medium of exchange or speculative assets? *Journal of International Financial Markets, Institutions and Money* 54, 177–189.
- Bhambhwani, Siddharth M.; Delikouras, Stefanos, Korniotis, George M. (2021). Do Fundamentals Drive Cryptocurrency Prices? SSRN working paper.
- Biais, B., C. Bisiere, M. Bouvard, C. Casamatta, and A. J. Menkveld (2018). Equilibrium Bitcoin pricing. *Working paper*.
- Bonaparte, Yosef. (2021A). “Time Horizon, Saving Motive and Stock Market Participation.” *SSRN working paper* 3833475.
- Bonaparte, Yosef. (2021B). “Introducing the Cryptocurrency VIX: CVIX.” *SSRN working paper* 3948898
- Bonaparte, Yosef. (2022). “Blockchain Technology and Crypto Financial Performance.” *Work in Progress*.
- Catalini, C. and J. S. Gans (2016). Some simple economics of the blockchain. *NBER Working Paper No. 22952*.
- Cheah, E.-T. and J. Fry (2015). Speculative bubbles in Bitcoin markets? An empirical investigation into the fundamental value of Bitcoin. *Economics Letters* 130, 32–36.
- Chiu, J. and T. V. Koepl (2019). Blockchain-based settlement for asset trading. *The Review of Financial Studies* 32 (5), 1716–1753.
- Choi, H., & Varian, H. (2012). Predicting the present with Google Trends. *Economic Record*, 88(s1), 2-9.
- Cong, L. W. and Z. He (2019). Blockchain disruption and smart contracts. *The Review of Financial Studies* 32 (5), 1754–1797.
- Cong, L. W., Y. Li, and N. Wang (2018). Tokenomics: Dynamic adoption and valuation. *Columbia Business School Research Paper No. 18-46* . 42
- Curcuro, Stephanie, John Heaton, Deborah Lucas, and Damien Moore, 2010, “Heterogeneity and Portfolio Choice: Theory and Evidence,” in *Handbook of Financial Econometrics: Tools and Techniques*, 337-382.
- Da, Z., Engelberg, J., and Gao, P. (2011). In search of attention. *The Journal of Finance*, 66(5), 1461-1499.
- Dimpfl, T., & Jank, S. (2016). Can internet search queries help to predict stock market volatility? *European Financial Management*, 22(2), 171-192.
- Dzielinski, M. (2012). Measuring economic uncertainty and its impact on the stock market. *Finance Research Letters*, 9(3), 167-175.
- Easley, D., M. O’Hara, and S. Basu (2019). From mining to markets: The evolution of Bitcoin transaction fees. *Journal of Financial Economics*, V(134) issue 1 pages: 91-109.
- Foley, S., J. R. Karlsen, and T. J. Putn, n’s (2019). Sex, drugs, and bitcoin: How much illegal activity is financed through cryptocurrencies? *The Review of Financial Studies* 32 (5), 1798–1853.

- Ginsberg, J., Mohebbi, M. H., Patel, R. S., Brammer, L., Smolinski, M. S., & Brilliant, L. (2009). Detecting influenza epidemics using search engine query data. *Nature*, 457(7232), 1012-1014.
- Giudici, G., Milne, A. and Vinogradov, D. (2020). Cryptocurrencies: market analysis and perspectives. *Journal of Industrial and Business Economics* 47, 1–18.
- Houy, N. (2014). The economics of Bitcoin transaction fees. *Working paper*.
- Huberman, G. and S. Kandel (1987). Mean-variance spanning. *The Journal of Finance* 42 (4), 873–888.
- Huberman, G., J. Leshno, and C. C. Moallemi (2017). Monopoly without a monopolist: An economic analysis of the bitcoin payment system. *Columbia Business School Research Paper* No. 17-92.
- Iyidogan, E. (2019). An equilibrium model of blockchain-based cryptocurrencies. *Working paper*
- Lemieux, J., & Peterson, R. A. (2011). Purchase deadline as a moderator of the effects of price uncertainty on search duration. *Journal of Economic Psychology*, 32(1), 33-44.
- Li, Q.-L., J.-Y. Ma, and Y.-X. Chang (2018). Blockchain queue theory. In *International Conference on Computational Social Networks*, pp. 25–40. Springer.
- Liu, Y. and A. Tsyvinski (2018). Risks and returns of cryptocurrency. *NBER Working Paper* No. 24877.
- Liu, Y., A. Tsyvinski, and X. Wu (2019). Common risk factors in cryptocurrency. *NBER Working Paper* No. 258882.
- Makarov, I. and A. Schoar (2019). Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics*, V(135), Issue 2, February 2020, Pages 293-319
- Malinova, K. and A. Park (2017). Market design with blockchain technology. *Working paper*.
- Nakamoto, S. (2008). Bitcoin: A peer-to-peer electronic cash system. *Online Publication*.
- Narayanan, A., J. Bonneau, E. Felten, A. Miller, and S. Goldfeder (2016). Bitcoin and cryptocurrency technologies: A comprehensive introduction. *Princeton University Press*.
- Pagano, M. S. and J. Sedunov (2019). Bitcoin and the demand for money: Is Bitcoin more than a speculative asset? *Working paper*.
- Pagnotta, E. and A. Buraschi (2018). An equilibrium valuation of Bitcoin and decentralized network assets. *Working paper*.
- Prat, J. and B. Walter (2018). An equilibrium model of the market for bitcoin mining. *Working paper*.
- Saleh, F. (2018). Blockchain without waste: Proof-of-stake. *Working paper*.
- Selgin, G. (2015). Synthetic commodity money. *Journal of Financial Stability* 17, 92–99.
- Sockin, M. and W. Xiong (2018). A model of cryptocurrencies. Unpublished manuscript, *Princeton University*.
- Tinn, K. (2017). Blockchain and the future of optimal financing contracts. *Working paper*.
- Yermack, D. (2015). Is Bitcoin a real currency? An economic appraisal. In *Handbook of Digital Currency*, pp. 31–43. Elsevier.
- Yermack, D. (2017). Corporate governance and blockchains. *Review of Finance* 21 (1), 7–31.
- Vlastakis, N., and Markellos, R. N. (2012). Information demand and stock market volatility. *Journal of Banking & Finance*, 36(6), 1808-1821.
- Vozlyublennaiia, N. (2014). Investor attention, index performance, and return predictability. *Journal of Banking & Finance*, 41, 17-35.

FIGURE 1: Crypto Regulation Sentiment Index - CRSX

This figure depicts the CRSX between 1/2004 and 3/2022. The blue line measures the Global CRSX while the black curve reports the US CRSX.

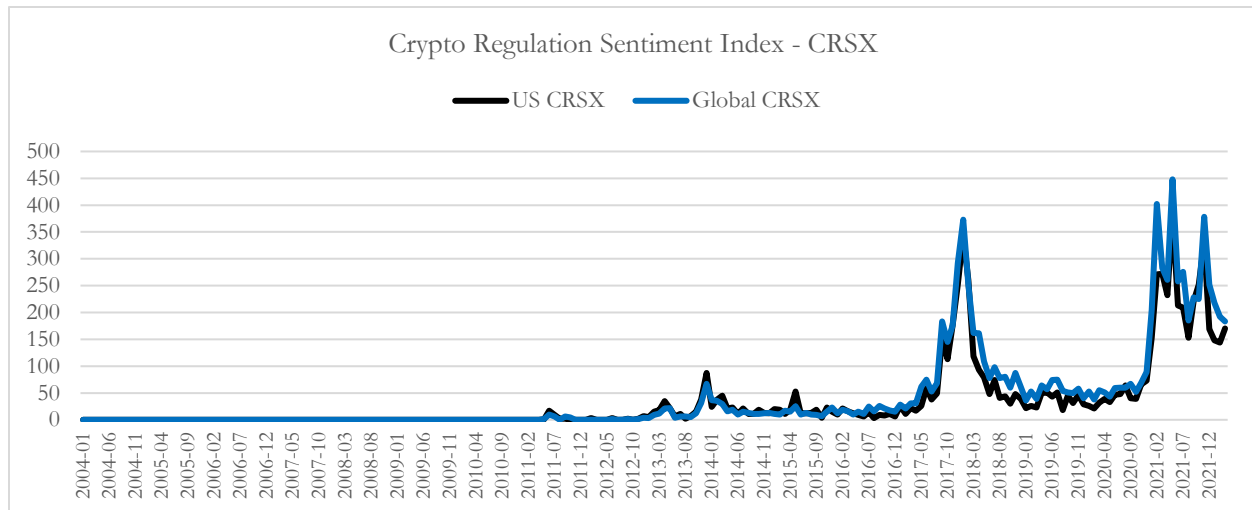


FIGURE 2: Bitcoin prices and Crypto Regulation Sentiment Index - CRSX

This figure depicts the CRSX the normalized price of Bitcoin between 1/2014 and 3/2022. The dark blue line measures the price of Bitcoin whereas the light blue depicts the Global CRSX

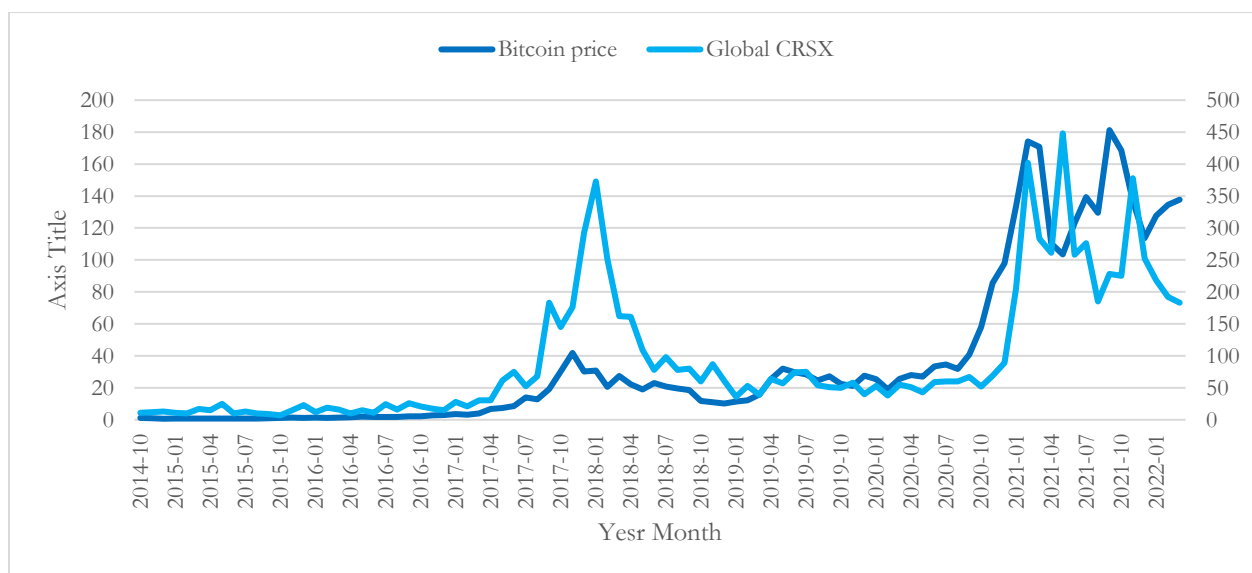


TABLE 1: summary statistics of key coin's characteristics

This table reports the summary statistics of each coin key characteristics. In column (1) we report the name of the coin; column (2) the age in months; column (3) the market cap in billions of dollars as of 4/10/2022; and in column (4) the type of the consensus mechanism where we assign a value of 1 for proof-of-work consensus mechanism, and 0 otherwise (proof-of-stake and others). The definitions of all variables are available in Appendix A.

Coin	Age (month)	Market Cap*	Consensus Mechanism
Bitcoin	90	810	1
Ethereum	79	390	1
BNB	56	69.8	0
Cardano	54	35	0
Solano	42	36.6	0
Terra	52	32.2	0
Dogecoin	90	20.8	1

TABLE 2: Summary Statistics of key variables

This table reports the summary statistics of our key variables. In Particular, we report the summary statistics (mean, standard deviation (Std. Dev), percentiles (p25th, p50th and p75th) and number of observation “N”). The definitions of all variables are available in Appendix A.

Variable	Mean	Std. Dev	Percentile:			N
			p25	P50	P75	
US CRSX	92.356	94.773	26	48	151	463
Global CRSX/100	1.156	1.074	0.43	0.64	1.85	463
Return %	2.801	56.195	-0.157	0.015	0.277	463
Volatility %	0.089	0.189	0.034	0.048	0.073	374
Log volume	21.993	5.734	20.906	23.121	25.798	365
Age	36.095	23.088	17	34	51	463
Market Cap	194.268	337.702	1.19	20.55385	100.01	463
Consensus Mechanism	0.559	0.497	0	1	1	463

TABLE 3: Global and US CRSX Time Series Analyses

This table reports the estimation results for key properties of the global and US CRSX. The dependent variable in regressions (1) and (2) is global CRSX and the key independent variable is the US CRSX. The dependent variable in regression (3) is global CRSX and the independent variable is lag CRSX. In regressions (1) and (3) we omitted the constant. The dependent variable in regression (4) is the US CRSX and the key independent variable is lag US CRSX. The number in parentheses measures the *t*-statistics and we employ OLS regression with robust standard error. Definition of key variables are in Appendix A.

VARIABLES	Dependent variable:			
	Global CRSX		US CRSX	
	(1)	(2)	(3)	(4)
US CRSX	1.178 (460.49)	1.111 (136.23)		
Lag Global CRSX			0.971 (63.80)	
Lag US CRSX				0.974 (61.87)
Constant	Omitted	0.126 (7.25)	0.036 (3.57)	0.027 (3.14)
Observations	411	411	410	410
R-squared	0.976	0.957	0.944	0.948

TABLE 4: Regulation sentiment and crypto prices

This table reports the estimation results on the impact of CRSX over crypto prices. The dependent variable in regressions (1), (2) and (5) is crypto return at time t (concurrent) and in regressions (4) and (5) crypto return with one month forward and two months forward, respectively. The key independent variable is global CRSX and the market cap, trading age of the crypto as well as the type of the consensus mechanism (1 if POF and 0 otherwise). We employed three interaction terms with CRSX: with market age (CRSX*Age); market cap (CRSX*MC) and consensus mechanism (CRSX*CM). In all regressions, we control for year and month fixed effects, weighted average by coin cap and cluster by coin. The number in parentheses measures the t -statistics and we employ OLS regression with robust standard error. Definition of key variables are in Appendix A.

VARIABLES	Dependent variable crypto return at time:				
	t	t	t+1	t+2	t
	(1)	(2)	(3)	(4)	(5)
Global CRSX	0.009 (0.22)	0.105 (0.85)	-0.313 (-1.11)	0.172 (2.35)	0.043 (0.31)
Market Cap (MC)	-0.000 (-0.73)	-0.000 (-0.81)	-0.002 (-1.11)	-0.001 (-1.59)	-0.000 (-3.42)
Age	-0.007 (-0.70)	-0.006 (-0.65)	0.010 (0.13)	0.041 (2.15)	0.001 (0.11)
Consensus Mechanism (CM)	-0.008 (-0.03)	-0.021 (-0.08)	-3.480 (-0.92)	-1.099 (-2.07)	-0.190 (-0.68)
CRSX*Age					-0.003 (-1.18)
CRSX*CM					0.075 (0.75)
SRSX*MC					0.000 (1.15)
Global CRSX ²		-0.021 (-1.01)			
Constant	0.034 (0.07)	0.061 (0.14)	5.234 (0.91)	1.373 (1.53)	0.322 (0.76)
Year/Month FE	Yes	Yes	Yes	Yes	Yes
Observations	446	446	445	444	446
R-squared	0.007	0.007	0.009	0.260	0.007

TABLE 5: Regulation sentiment and crypto volatility

This table reports the estimation results on the impact of CRSX over crypto prices. The dependent variable in regressions (1), (2) and (5) is crypto volatility (standard deviation of daily return) at time t (concurrent) and in regressions (4) and (5) crypto volatility with one month forward and two months forward, respectively. The key independent variable is global CRSX and the market cap, trading age of the crypto as well as the type of the consensus mechanism (1 if POF and 0 otherwise). We employed three interaction terms with CRSX: with market age (CRSX*Age); market cap (CRSX*MC) and consensus mechanism (CRSX*CM). In all regressions, we control for year and month fixed effects, weighted average by coin cap and cluster by coin. The number in parentheses measures the t -statistics and we employ OLS regression with robust standard error. Definition of key variables are in Appendix A.

VARIABLES	Dependent variable crypto volatility at time:				
	t	t	t+1	t+2	t
	(1)	(2)	(3)	(4)	(5)
Global CRSX	0.013 (5.07)	0.019 (2.15)	0.008 (1.84)	0.014 (2.52)	-0.028 (-2.32)
Market Cap (MC)	0.000 (0.36)	0.000 (0.33)	-0.000 (-1.82)	-0.000 (-2.89)	0.000 (0.15)
Age	-0.002 (-0.98)	-0.002 (-0.95)	-0.000 (-0.42)	0.002 (2.65)	-0.001 (-0.67)
Consensus Mechanism (CM)	-0.048 (-0.62)	-0.048 (-0.63)	0.041 (2.99)	-0.054 (-2.38)	-0.178 (-1.94)
CRSX*Age					-0.000 (-1.65)
CRSX*CM					0.060 (3.43)
SRSX*MC					0.000 (0.35)
Global CRSX ²		-0.001 (-0.66)			
Constant	0.066 (0.60)	0.067 (0.61)	0.019 (0.45)	0.167 (3.22)	0.202 (1.77)
Year/Month FE	Yes	Yes	Yes	Yes	Yes
Observations	374	374	356	359	374
R-squared	0.260	0.260	0.224	0.270	0.302

TABLE 6: Regulation sentiment and crypto log trading volume

This table reports the estimation results on the impact of CRSX over crypto prices. The dependent variable in regressions (1), (2) and (5) is crypto log trading volume at time t (concurrent) and in regressions (4) and (5) crypto log trading volume with one month forward and two months forward, respectively. The key independent variable is global CRSX and the market cap, trading age of the crypto as well as the type of the consensus mechanism (1 if POF and 0 otherwise). We employed three interaction terms with CRSX: with market age (CRSX*Age); market cap (CRSX*MC) and consensus mechanism (CRSX*CM). In all regressions, we control for year and month fixed effects, weighted average by coin cap and cluster by coin. The number in parentheses measures the t -statistics and we employ OLS regression with robust standard error. Definition of key variables are in Appendix A.

VARIABLES	Dependent variable crypto log trading volume at time:				
	t	t	t+1	t+2	t
	(1)	(2)	(3)	(4)	(5)
Global CRSX	0.358 (1.59)	1.004 (1.79)	0.786 (5.23)	-0.767 (-1.19)	2.790 (7.19)
Market Cap (MC)	-0.001 (-0.42)	-0.001 (-0.45)	0.001 (1.84)	0.001 (0.51)	0.000 (0.38)
Age	0.154 (0.88)	0.159 (0.90)	0.256 (3.13)	-0.068 (-0.52)	0.240 (1.33)
Consensus Mechanism (CM)	6.215 (0.88)	6.138 (0.86)	-11.892 (-4.18)	0.006 (0.00)	6.804 (1.02)
CRSX*Age					-0.020 (-3.01)
CRSX*CM					-0.784 (-1.08)
SRSX*MC					-0.001 (-1.64)
Global CRSX ²		-0.144 (-1.79)			
Constant	15.401 (1.56)	15.575 (1.57)	35.044 (9.35)	20.948 (3.48)	13.908 (1.58)
Year/Month FE	Yes	Yes	Yes	Yes	Yes
Observations	365	365	347	350	365
R-squared	0.799	0.800	0.312	0.207	0.824

TABLE 7: Regulation sentiment and crypto volatility and trading volume post 2020

This table reports the estimation results on the impact of CRSX over crypto return, volatility and volume for the subsamples before and after 1/2020. The dependent variable in regressions (1) and (2) is crypto return, while in regressions (3) and (4) is crypto volatility and in regressions (5) and (6) is crypto trading volume. The key independent variable is global CRSX, market cap, age and consensus mechanism of the crypto. In all regressions, we control for year and month fixed effects, weighted average by coin cap and cluster by coin. The number in parentheses measures the *t*-statistics and we employ OLS regression with robust standard error. Definition of key variables are in Appendix A.

VARIABLES	Dependent variable:					
	Return		Volatility		Log Volume	
	(1)	(2)	(3)	(4)	(5)	(6)
Global CRSX	0.006 (0.15)	0.008 (0.14)	-0.004 (-0.39)	0.012 (22.81)	0.029 (0.38)	0.949 (6.05)
Market Cap (MC)	-0.001 (-3.06)	0.000 (1.26)	-0.000 (-0.94)	-0.000 (-0.31)	0.000 (0.14)	-0.000 (-0.14)
Age	0.013 (0.80)	-0.013 (-2.79)	-0.000 (-0.18)	-0.001 (-0.80)	0.055 (0.43)	0.157 (1.37)
Consensus Mechanism (CM)	-0.270 (-0.74)	0.019 (0.11)	-0.050 (-0.79)	-0.117 (-1.98)	4.576 (0.96)	10.261 (2.33)
Constant	0.196 (0.31)	-0.299 (-1.02)	0.143 (1.48)	0.156 (1.83)	18.972 (3.25)	10.287 (1.67)
Year/Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Sample 2020	Before	After	Before	After	Before	After
Observations	187	259	186	188	177	188
R-squared	0.004	0.263	0.100	0.521	0.430	0.968

Appendix A

Definition of key variables

This table reports summary definition for variables used in our analysis. Specifically, we report the variable name that used in our analysis, definition and the source. Created variables are those who utilize information from one or more variables and recoded the information such that to be more suitable for common econometrics methodology and for empirical analysis. The data about Bitcoin and Ethereum is available in many sources, including Yahoo Finance.

Variable	Description	Source
US CRSX	US Cryptocurrency Regulation Sentiment Index	Created
Global CRSX	Global Cryptocurrency Regulation Sentiment Index	Created
Crypto return %	Monthly return of crypto=(end of month price-beginning of month return)/beginning of month price	Created
Crypto volatility %	standard deviation of the daily returns by month	Created
Log volume	log the sum of the daily volume of crypto	Created
Age	Monthly data that report the coin number of trading month	
Market cap	the value of the coin by month	https://coinmarketcap.com/
Consensus mechanism	A binary variable takes a value of 1 if the coin consensus mechanism is proof-of-work, 0 otherwise (proof-of-stake, etc.)	Created